

Research on improving views by controllable textual attributes

1. Background

From an author's perspective, increasing the attention of a paper measured by the number of views is a common objective. However under realistic constraints, many factors that influence visibility are not adjustable. Publication year is fixed due to its strict temporal relevance, while authors and target journals are largely constrained by research scope and academic positioning. As a result, the only controllable attributes available for improvement are the article's title, abstract, and pages.

This study therefore investigates how these controllable features relate to article views, with the aim of understanding whether and to what extent they can contribute to higher attention.

2. Data Preprocessing

2.1 Objective

The raw dataset contains heterogeneous formats and unstructured text, making it unsuitable for direct analysis. A comprehensive preprocessing pipeline is required.

2.2 Step 1: Data Extraction

The following variables are extracted from the dataset:

- Article title
- Abstract
- Number of views

2.3 Step 2: Text Preprocessing

For both titles and abstracts, the following preprocessing steps are applied:

- Convert all text to lowercase
- Remove punctuation
- Remove stop words that don't contribute to prediction
- Apply stemming to reduce words to their root forms
- Tokenize text into documents and individual words

3.LDA Topic Modeling

3.1 Objective

This study aims to assess whether pages and the wording of titles and abstracts that acted as proxies for research directions, has a positive influence on article views. Since titles and abstracts are unstructured text, they can't be directly incorporated into quantitative models. Topic modeling is therefore used to transform text into structured, numerical representations.

In addition, examining the distribution of topics helps identify potential sample imbalance that may affect subsequent analyses.

3.2 Step 1: Topic Representation via LDA

Separate Latent Dirichlet Allocation (LDA) models using the preprocessed text. Low-frequency components in the topic–word matrices are removed, and then the models are retrained to improve robustness.

- Each abstract is represented as a probability distribution over **20** topics
- Each title is represented as a probability distribution over **10** topics
- Each topic corresponds to a probability distribution over **50** words

3.3 Step 2: Interpretation of LDA Results

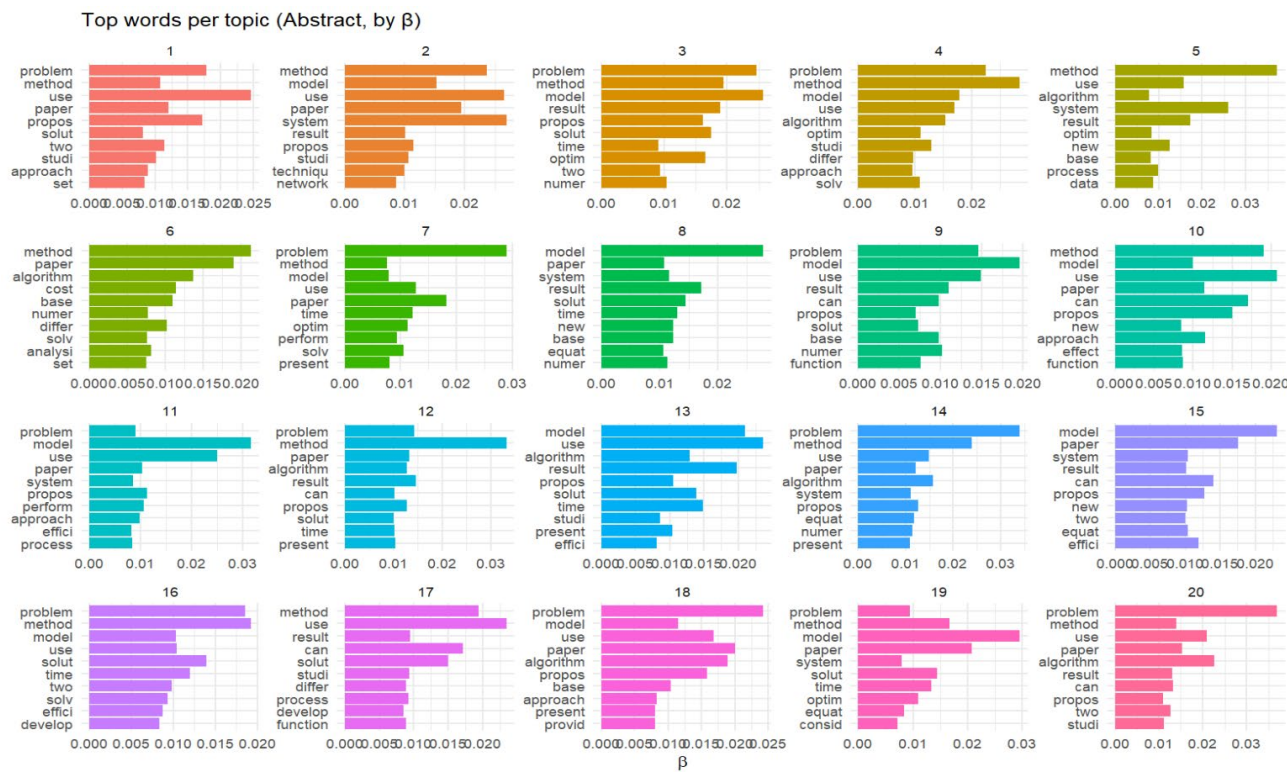


Figure 3.3.a Top Words per Topic of Abstract

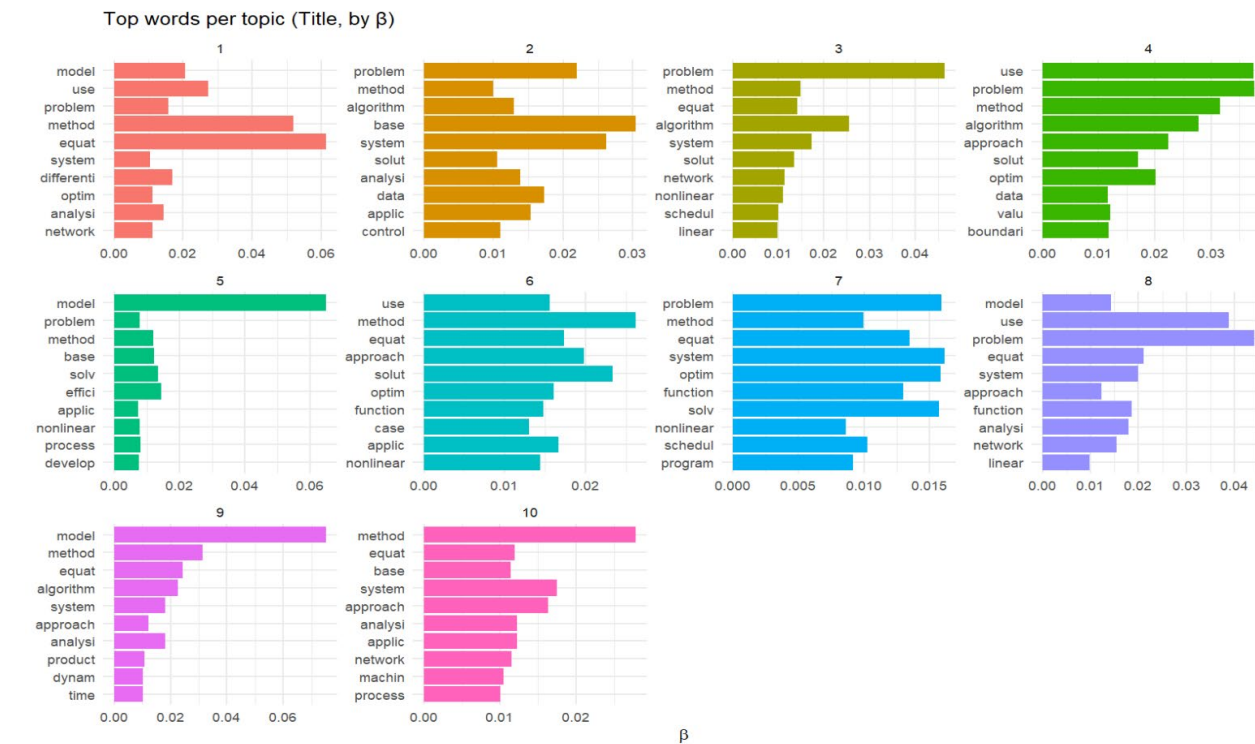


Figure 3.3.b Top Words per Topic of Title

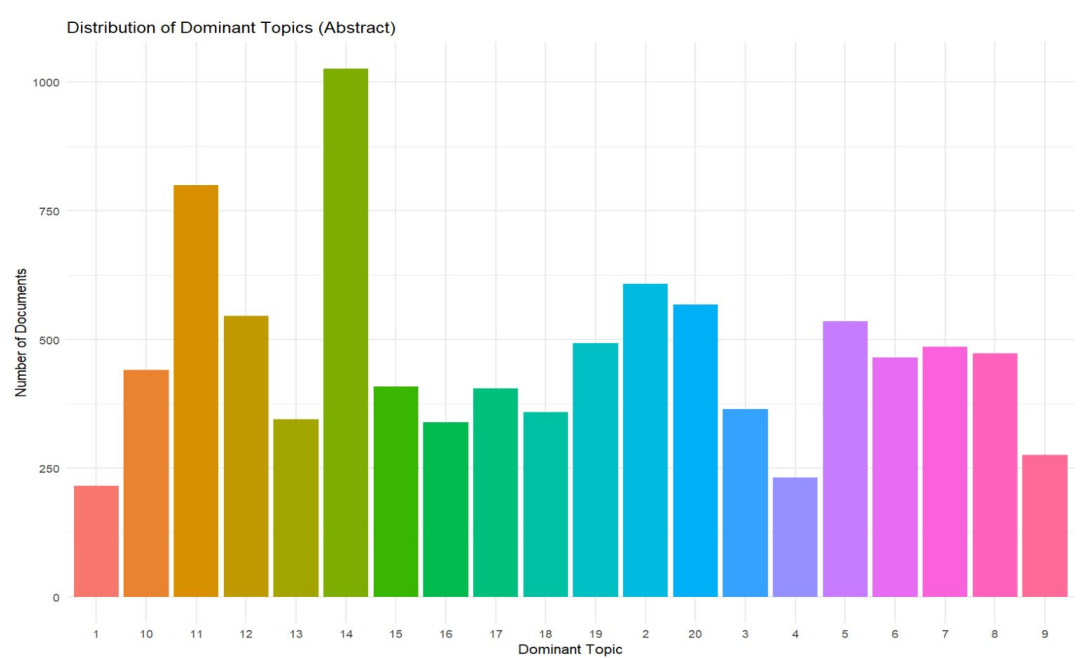


Figure 3.3.c Number of Dominant Topics of Abstract

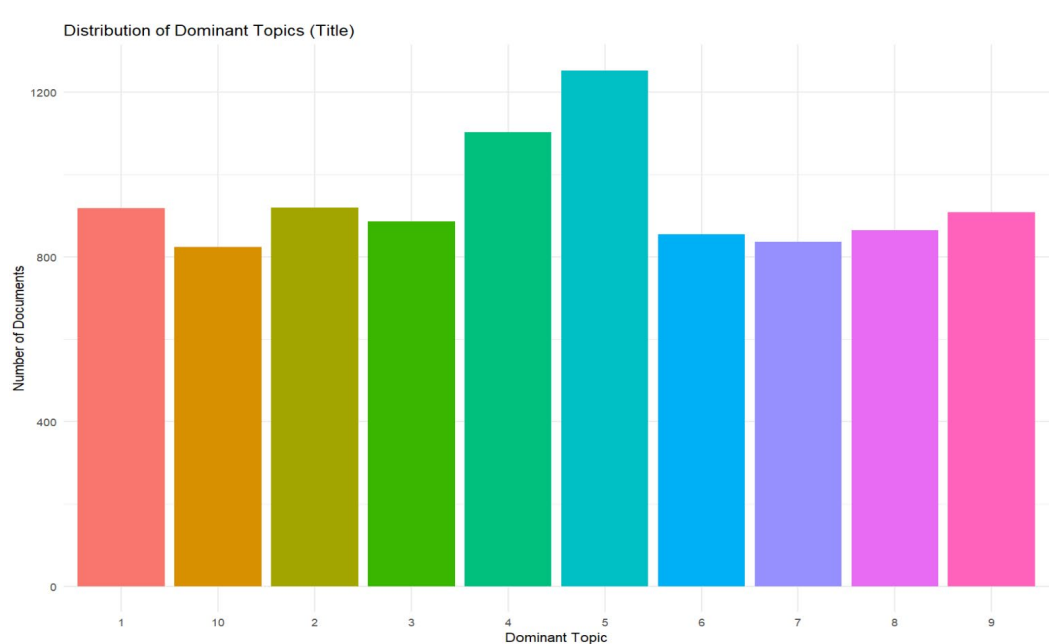


Figure 3.3.d Number of Dominant Topics of Titles

The LDA results demonstrate that both titles and abstracts can approximately distinguish between different research directions, with each topic characterized by a set of high-probability terms. For example, Topic 1 emphasizes the term *“use”*, whereas Topic 3 places greater emphasis on *“model”* in abstracts.

Abstracts exhibit more distinct topic–word structures than titles, suggesting that they contain richer semantic information for identifying underlying research directions. Certain topics, such as Topic 11 and Topic 14, attract a larger number of documents, indicating that some research directions are more prevalent in the dataset.

In contrast, topic–word distributions in titles are more compact, though still informative. The dominant topic distribution among titles is relatively balanced overall, despite number of dominants of Topics 4 and 5.

Overall, abstracts tend to highlight dominant themes, while titles describe research directions more concisely. LDA successfully captures meaningful thematic structure within the corpus and provides a reasonable approximation for grouping articles by research direction.

4.Bag of Words and TF-IDF Analysis

4.1 Objective

The LDA results suggest that terms such as “*use*”, “*method*”, “*model*”, and “*problem*” dominate many topics. To validate this observation from a global perspective, Bag of Words and TF-IDF analyses are conducted on all titles and abstracts.

4.2 Results and Interpretation

word<chr>	total_tf_idf<dbl>	total_tf<dbl>	total_idf<dbl>	n_docs<int>
equat	247.4088	129.15602	2068.827	1080
method	222.4730	170.53106	1749.454	1341
model	222.2416	159.36633	1662.283	1192
system	190.8344	107.07074	1461.503	820
algorithm	181.2687	96.50696	1342.982	715
optim	166.9459	82.25306	1254.331	618
analysi	163.1805	80.55847	1280.189	632
base	154.5128	97.91855	1338.120	848
network	148.7973	57.27527	1184.657	456
schedul	142.5496	46.06133	1055.320	341

Figure 4.2.a Dominant Words of Abstracts

word<chr>	total_tf_idf<dbl>	total_tf<dbl>	total_idf<dbl>	n_docs<int>
method	205.2228	157.30841	4893.514	3751
model	164.5829	118.02009	4827.873	3462
paper	144.8134	111.32087	6639.614	5104
solut	140.3735	75.48384	4461.299	2399
propos	137.3066	79.85604	5765.237	3353
algorithm	137.2834	73.08934	4045.850	2154
result	131.4484	85.17805	6080.284	3940
system	127.6274	71.60741	4167.066	2338
function	120.5076	55.04212	3853.292	1760
numer	120.0612	61.85975	4531.912	2335

Figure 4.2.b Dominant Words of Titles

Term popularity who closely related to dominant topic words and topic prevalence is well reflected by the Total_TF_IDF score. The results confirm the initial hypothesis that “*method*” and “*model*” are the most influential terms in the corpus.

Interestingly, although “*use*” appears frequently among top topic words in both titles and abstracts, its total TF-IDF score is relatively low likely due to its widespread presence across many minor topics.

5.Clustering Analysis

5.1 Objective

Although topic distributions are high-dimensional, it doesn't lead to effective separation between articles definitely. Before proceeding to predictive modeling, clustering analysis on all articles is used to evaluate whether topic proportions derived from titles and abstracts can distinguish different research directions.

If topic distributions exhibit weak separability, regression modelling of article views would not be appropriate.

5.2 Step 1: K-means Clustering

K-means clustering is performed using topic proportions from titles and abstracts as input features, producing five clusters. Each cluster is interpreted as representing a group of related research directions.

For each cluster, dominant title and abstract topics are examined to characterize its thematic focus. Principal Component Analysis (PCA) is then applied to reduce dimensionality and visualize cluster separation.

cluster <fctr>	topic_index <int>	mean_proportion <dbl>
1	1	0.00000000
1	2	0.00000000
1	3	0.00000000
1	4	0.00000000
1	5	0.00000000
2	1	0.10048239
2	4	0.10019776
2	6	0.10017611
2	3	0.10000942
2	7	0.09997689
3	5	0.10060368
3	9	0.10024834
3	3	0.10012951
3	10	0.10003699
3	7	0.09999621
4	5	0.10037427
4	2	0.10019660
4	10	0.10010417
4	8	0.10007782
4	9	0.10005349
5	3	0.10044465
5	4	0.10030957
5	7	0.10016927
5	8	0.10013115
5	2	0.09992584

Figure 5.2.a Top Topics of Abstract Clusters

cluster <fctr>	topic_index <int>	mean_proportion <dbl>
1	10	0.05019367
1	6	0.05014652
1	9	0.05011853
1	17	0.05011388
1	2	0.05004970
2	14	0.05037452
2	12	0.05027693
2	8	0.05019072
2	16	0.05016973
2	17	0.05013910
3	19	0.05068249
3	6	0.05059558
3	11	0.05058722
3	10	0.05043902
3	15	0.05037252
4	11	0.05030574
4	2	0.05029580
4	5	0.05014637
4	10	0.05014511
4	13	0.05009836
5	20	0.05056181
5	7	0.05042601
5	18	0.05036654
5	3	0.05035258
5	14	0.05022291

Figure 5.2.b Top Topics of Title Clusters

Top topics of each cluster is produced in the process. By identifying the top topics within each cluster, one can assess whether the clusters exhibit sufficiently strong separability. If separation of clusters is high enough to justify predictive modeling, the weights of an article on dominant topics of its cluster can be treated as continuous input features of a regression model.

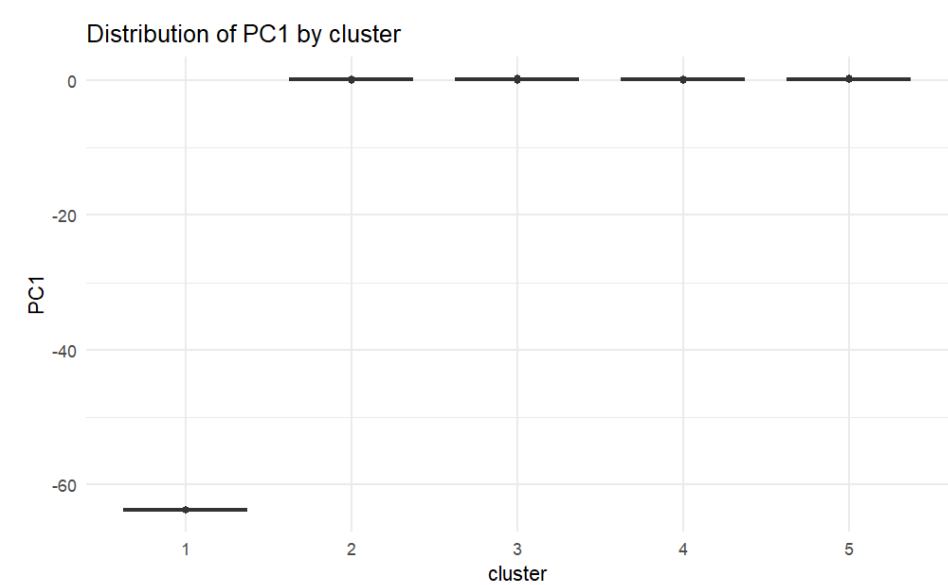


Figure 5.2.c Distribution of PC1 by Cluster

Obviously, PC1 is dominated by Cluster 1, indicating that it captures variation specific to a small subset of documents rather than general structural differences.

817	Random-tree Diameter and the Diameter-constrained MST	International Journal of Computer Mathematics	A minimum spanning tree (MST) with a small diameter is r...	13	51	1
932	Panel discussion	Applied Artificial Intelligence	This document is a summary of the panel discussion that w...	28	21	1
1156	How much should we reweigh?	Journal of the Operational Research Society	Suppose that in the final stage of manufacturing of, for inst...	4	6	1
1207	The intellectual journey of Russell Ackoff: from OR apostle t...	Journal of the Operational Research Society	Russell Ackoff has had a distinguished career in operational ...	14	61	1
1627	Retroreflection	Journal of Interdisciplinary Mathematics	Abstract With plane mirror combinations, we construct pat...	16	24	1
2408	Lawtistics	Journal of Interdisciplinary Mathematics	Abstract In this work the author proffers a novel approach t...	31	9	1
2753	Pragmatic inquiry	Journal of the Operational Research Society	Inquiry into how people might act to solve a problem, whic...	9	122	1
3041	Remarks on symmetries of 2D-quasicrystals	International Journal of Computer Mathematics	Abstract The paper presents a strict mathematical proof of ...	10	47	1
3082	Heisenberg superalgebras	International Journal of Computer Mathematics	Heisenberg algebras are the only Lie algebras ($\mathfrak{g}$, $[\cdot, \cdot]$) which ...	12	138	1
3221	λ credibility	Journal of Interdisciplinary Mathematics	Abstract This paper extends credibility measure in uncertai...	14	26	1
4136	P-disjunctive ω -languages	International Journal of Computer Mathematics	An ω -language over a finite alphabet X is a set of infinite se...	7	55	1
4266	On L-fuzzy Chu correspondences	International Journal of Computer Mathematics	In this paper, we focus on the framework of Chu correspon...	11	67	1
4277	GEM and the light unflavored non-scalar mesons	Journal of Interdisciplinary Mathematics	Abstract In a recent article in The Journal of Interdisciplar...	15	12	1
5110	Topology-preserving hexagonal thinning	International Journal of Computer Mathematics	Thinning is a well-known technique for producing skeleton...	11	82	1
5701	Expectation-maximization for Bayes-adaptive POMDPs	Journal of the Operational Research Society	Bayes-adaptive POMDPs (BAPOMDPs) are partially observa...	19	49	1
5954	The unruly $\sin(\sqrt{e}/7)$	Journal of Interdisciplinary Mathematics	We give the first (to the best of our knowledge) closed-form...	6	31	1
6001	The unruly $\sin(\pi/7)$	Journal of Interdisciplinary Mathematics	We give the first (to the best of our knowledge) closed-form...	6	31	1
7637	On cocycles	Journal of Interdisciplinary Mathematics	Time dependent physical systems without impulse can be d...	9	18	1
7958	Gesture and Speech Recognizing Helper Bot	Applied Artificial Intelligence	In industries, difficult work is being decreased everywhere s...	11	401	1

Table 5.2.d Titles of Cluster 1

Notably, many titles in Cluster 1 contain words that were excluded from the LDA topic distribution due to low frequency, which may explain the abnormal distribution along PC1.

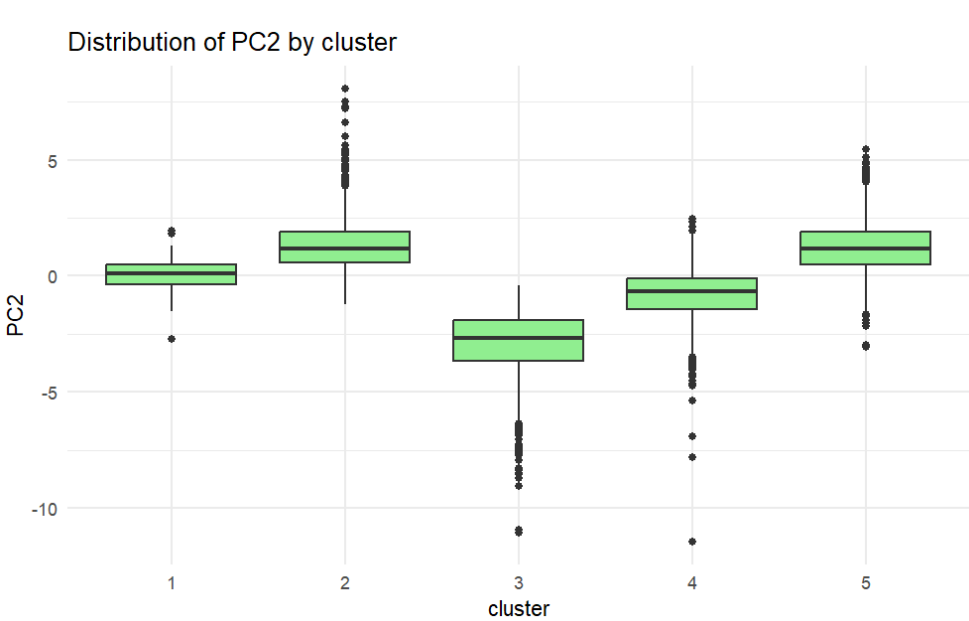


Figure 5.2.e Distribution of PC2 by Cluster

PC2 provides better cluster separation, though substantial overlap remains. Overall, K-means clustering reveals weak and unstable cluster structure.

5.3 Step 2: Hierarchical Clustering

Hierarchical clustering is conducted using the same topic proportion features, again producing five clusters. Representative topics for titles and abstracts are identified for each cluster.

cluster <fctr>	topic_index <int>	mean_proportion <dbl>
1	3	0.10066910
1	5	0.10038092
1	9	0.10032620
1	10	0.10008955
1	7	0.10001994
2	5	0.10035415
2	2	0.10016769
2	10	0.10005967
2	9	0.10004846
2	8	0.10003961
3	3	0.10035900
3	4	0.10027790
3	7	0.10015249
3	8	0.10011701
3	2	0.09993344
4	1	0.10068658
4	4	0.10027378
4	6	0.10018809
4	3	0.10003496
4	9	0.09999120
5	1	0.00000000
5	2	0.00000000
5	3	0.00000000
5	4	0.00000000
5	5	0.00000000

Figure 5.3.a Top Topics of Abstract Clusters

cluster <fctr>	topic_index <int>	mean_proportion <dbl>
1	6	0.05107212
1	19	0.05102750
1	10	0.05034606
1	7	0.05030577
1	11	0.05028388
2	11	0.05029590
2	2	0.05016890
2	10	0.05015250
2	15	0.05008714
2	13	0.05007462
3	20	0.05045350
3	7	0.05033872
3	3	0.05031358
3	18	0.05028594
3	14	0.05022891
4	14	0.05046502
4	12	0.05033882
4	16	0.05021489
4	8	0.05020863
4	5	0.05015175
5	10	0.05019367
5	6	0.05014652
5	9	0.05011853
5	17	0.05011388
5	2	0.05004970

Figure 5.3.b Top Topics of Title Clusters

Top topics of each cluster is produced in the process.

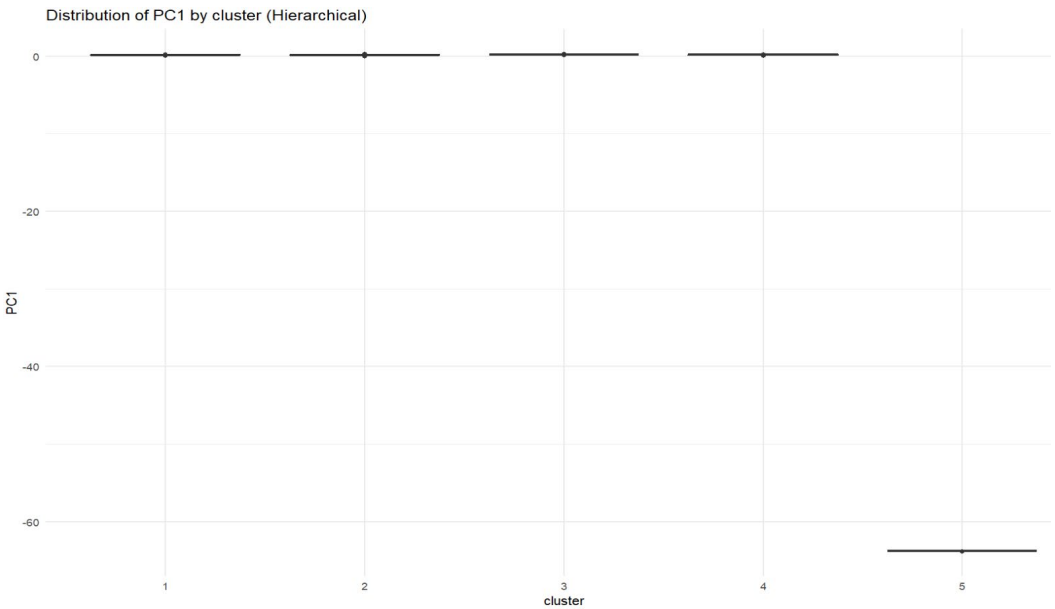


Figure 5.3.c Distribution of PC1 by Cluster

After dimensionality reduction with PCA, now PC1 is dominated by Cluster 5, indicating variation specific to a subset of articles.

	title	journal	abstract	pages	views	cluster
307	Medal awards 2000	Journal of the Operational Research Society	"Medal awards 2000." Journal of the Operational Research S...	2	4	5
427	John Anthony Hardinge Giffard, 3rd Earl of Halsbury	Journal of the Operational Research Society	"John Anthony Hardinge Giffard, 3rd Earl of Halsbury." Jour...	2	2	5
817	Random-tree Diameter and the Diameter-constrained MST	International Journal of Computer Mathematics	A minimum spanning tree (MST) with a small diameter is r...	13	51	5
932	Panel discussion	Applied Artificial Intelligence	This document is a summary of the panel discussion that w...	28	21	5
1156	How much should we reweigh?	Journal of the Operational Research Society	Suppose that in the final stage of manufacturing of, for inst...	4	6	5
1207	The intellectual journey of Russell Ackoff: from OR apostle t...	Journal of the Operational Research Society	Russell Ackoff has had a distinguished career in operational ...	14	61	5
1627	Retroreflection	Journal of Interdisciplinary Mathematics	Abstract With plane mirror combinations, we construct pat...	16	24	5
2408	Lawtistics	Journal of Interdisciplinary Mathematics	Abstract In this work the author proffers a novel approach t...	31	9	5
2753	Pragmatic inquiry	Journal of the Operational Research Society	Inquiry into how people might act to solve a problem, whic...	9	122	5
3041	Remarks on symmetries of 2D-quasicrystals	International Journal of Computer Mathematics	Abstract The paper presents a strict mathematical proof of ...	10	47	5
3082	Heisenberg superalgebras	International Journal of Computer Mathematics	Heisenberg algebras are the only Lie algebras $(\mathfrak{g}, [,])$ which ...	12	138	5
3221	λ credibility	Journal of Interdisciplinary Mathematics	Abstract This paper extends credibility measure in uncertai...	14	26	5
4136	P-disjunctive ω -languages	International Journal of Computer Mathematics	An ω -language over a finite alphabet X is a set of infinite se...	7	55	5
4266	On L-fuzzy Chu correspondences	International Journal of Computer Mathematics	In this paper, we focus on the framework of Chu correspon...	11	67	5
4277	GEM and the light unflavored non-scalar mesons	Journal of Interdisciplinary Mathematics	Abstract In a recent article in The Journal of Interdisciplar...	15	12	5
5110	Topology-preserving hexagonal thinning	International Journal of Computer Mathematics	Thinning is a well-known technique for producing skeleton...	11	82	5
5701	Expectation-maximization for Bayes-adaptive POMDPs	Journal of the Operational Research Society	Bayes-adaptive POMDPs (BAPOMDPs) are partially observa...	19	49	5
5954	The unruly $\sin(\frac{1}{e}/7)$	Journal of Interdisciplinary Mathematics	We give the first (to the best of our knowledge) closed-form...	6	31	5
6001	The unruly $\sin(\frac{1}{\pi}/7)$	Journal of Interdisciplinary Mathematics	We give the first (to the best of our knowledge) closed-form...	6	31	5

Figure 5.3.d Titles of Cluster 5

Interestingly, titles in Cluster 5 largely overlap with those in Cluster 1 identified by K-means clustering.

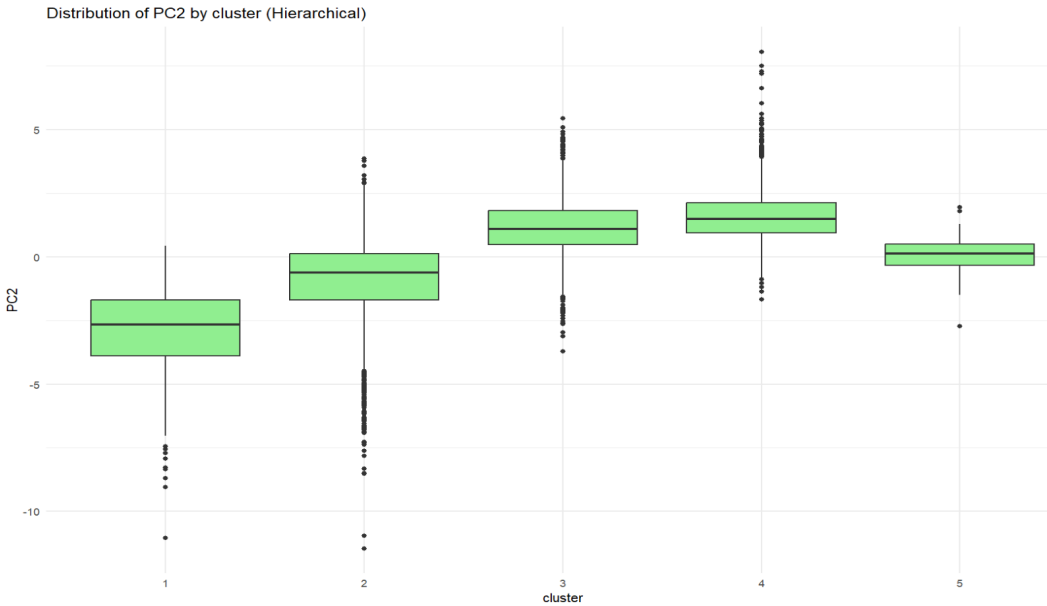


Figure 5.3.e Distribution of PC2 by Cluster

PC2 separates Clusters 1–4 more clearly, suggesting meaningful differences in topic composition. Overall, hierarchical clustering captures variation more effectively than K-means.

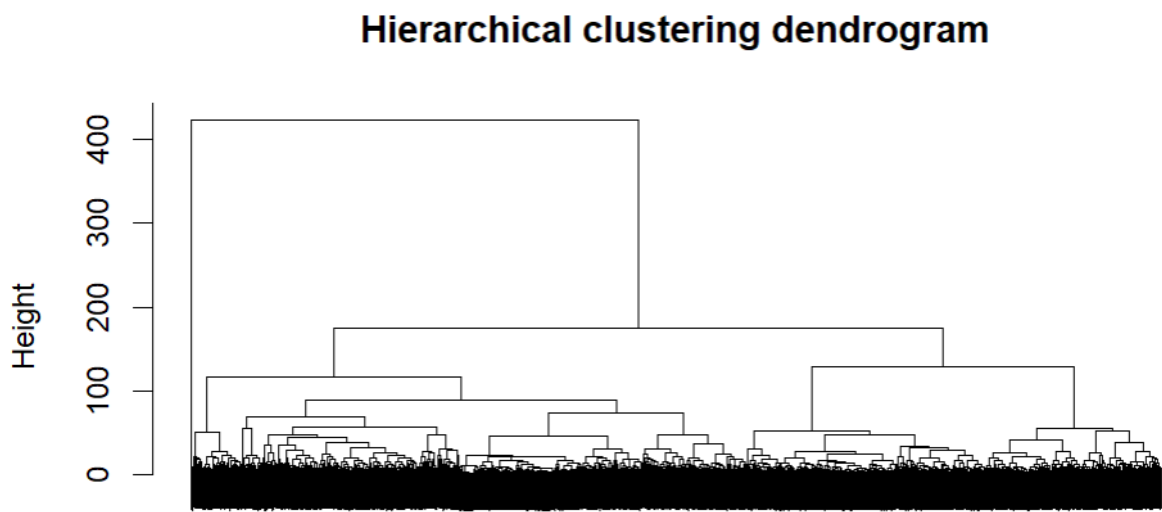


Figure 5.3.f Dendrogram

The dendrogram reveals a small number of broad clusters at high linkage distances, while separation at lower distances is weak. This pattern suggests a continuous topic structure rather than well-separated research categories.

5.4 Step 3: Summary

Although hierarchical clustering exhibits clearer structure than K-means, topic distributions like titles' remain highly overlapping across clusters. This indicates that topic composition alone is insufficient to reliably discriminate research directions.

6.Association Analysis

6.1 Objective

Given the limited discriminative power of topic distributions observed in clustering analysis, regression-based modeling is abandoned. Association analysis is employed instead to explore whether specific combinations of topic distributions, pages, and views tend to co-occur. Association analysis doesn't assume an explicit functional relationship between variables, making it more suitable for uncovering sparse and nonlinear patterns than regression.

6.2 Step 1: Discretization and Encoding

Topic proportions, pages, and views are continuous variables, making them unsuitable for association rule mining. Therefore these variables are then discretized and transformed into binary features using binning and one-hot encoding.

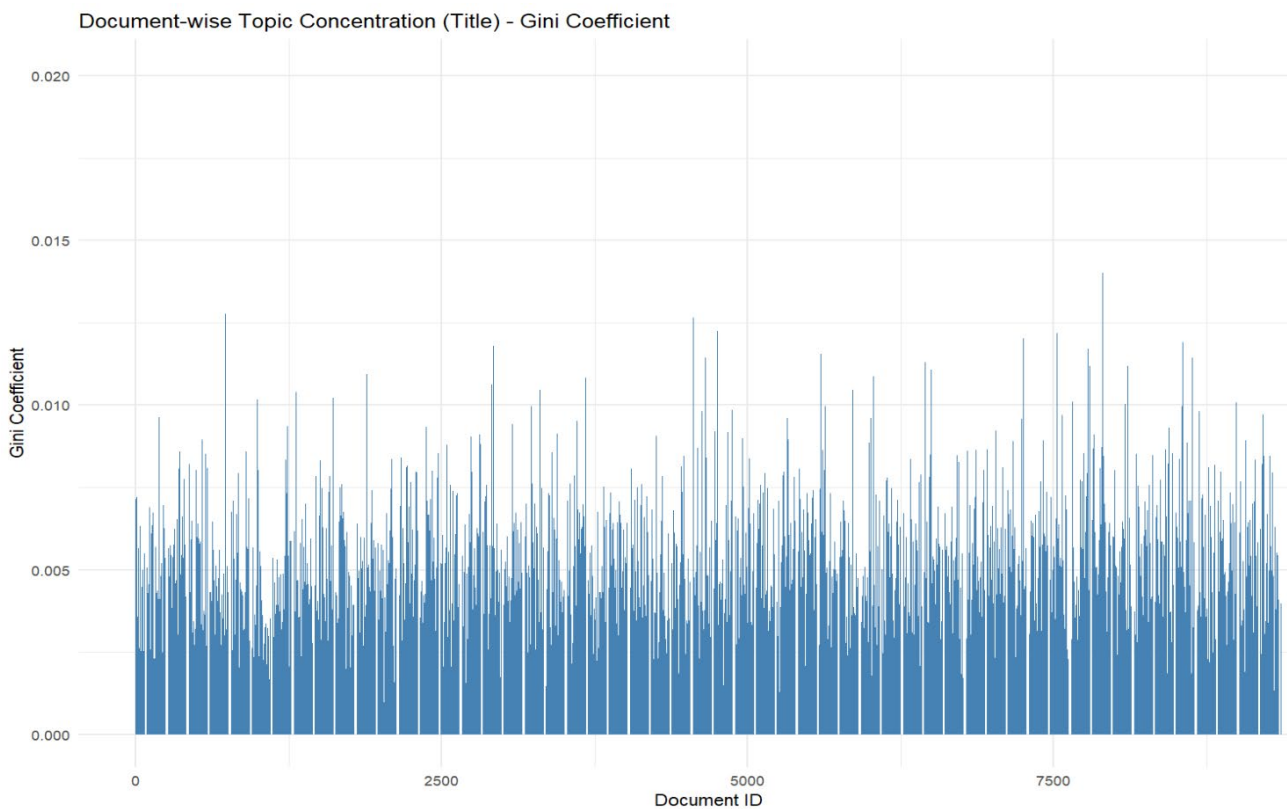


Figure 6.2.a Gini of Topics per Title

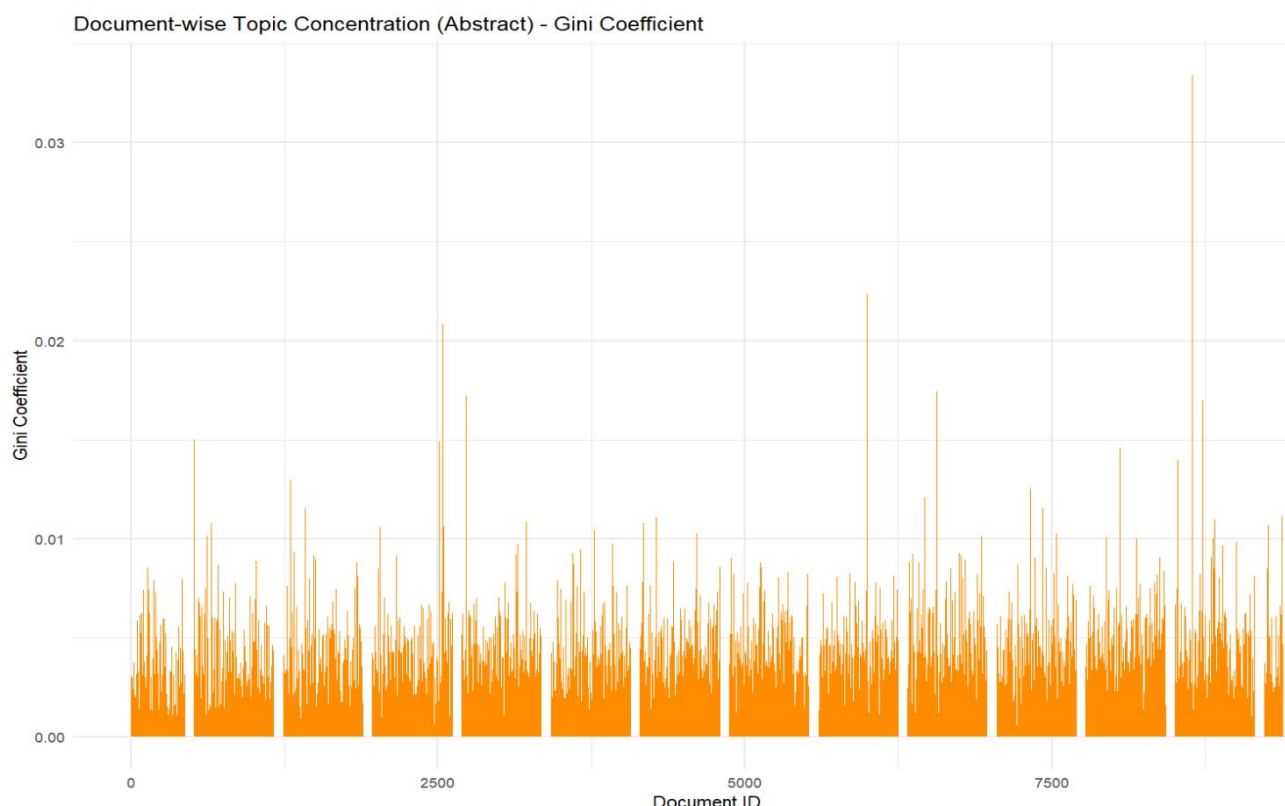


Figure 6.2.b Gini of Topics per Abstract

To identify dominant topics before one-hot encoding, the Gini coefficient of topic distributions is computed for each article. Most titles and abstracts exhibit Gini coefficients around **0.01**, indicating nearly uniform topic distributions and weak dominance.

To increase discriminative power, topics are ranked by proportion, and the smallest set of topics whose cumulative proportion reaches **50%** is selected as dominant. These dominant topics are then one-hot encoded.

```
> which(data_features$Abstract_11==1)
integer(0)
> which(data_features$Abstract_11==1)
integer(0)
> which(data_features$Abstract_12==1)
integer(0)
> which(data_features$Abstract_13==1)
integer(0)
> which(data_features$Abstract_14==1)
integer(0)
> which(data_features$Abstract_15==1)
integer(0)
> which(data_features$Abstract_16==1)
integer(0)
> which(data_features$Abstract_17==1)
integer(0)
> which(data_features$Abstract_18==1)
integer(0)
> which(data_features$Abstract_19==1)
integer(0)
> which(data_features$Abstract_20==1)
integer(0)
```

Figure 6.2.c Abandoned Abstracts

After encoding, abstract topics **11–20** contain only zero values and are removed to reduce dimensionality, effectively excluding them from further analysis.

Article views and pages are then discretized into five equal-width bins (bin_1 to bin_5), where higher bin indices correspond to higher values.

6.3 Step 2: Association Rule Mining

With all variables encoded, association rules among topic distributions, pages, and views are mined using the Apriori algorithm with pruning. Each article is treated as a transaction, and support, confidence, and lift are computed for rules involving high-view bins.

rules <chr>	support <dbl>	confidence <dbl>	lift <dbl>
1 {Abstract_1,Abstract_5,Abstract_8,pages=bin_2} => {views=bin_4}	0.000106553	0.02325581	218.2558
2 {Abstract_1,Abstract_5,Abstract_6,pages=bin_2} => {views=bin_4}	0.000106553	0.03030303	284.3939
3 {Abstract_1,Abstract_5,pages=bin_2,Title_6} => {views=bin_4}	0.000106553	0.02272727	213.2955
4 {Abstract_1,Abstract_6,Abstract_8,pages=bin_2} => {views=bin_4}	0.000106553	0.02325581	218.2558
5 {Abstract_1,Abstract_6,pages=bin_2,Title_6} => {views=bin_4}	0.000106553	0.02127660	199.6809
6 {Abstract_1,Abstract_6,pages=bin_2,Title_3} => {views=bin_4}	0.000106553	0.02000000	187.7000

Figure 6.3.a Analysis of Rules with High Views

The resulting rules exhibit extremely low support and confidence (on the order of 1×10^{-4}), indicating that such patterns are exceedingly rare. Rules associated with the highest view bin (bin_5) are removed due to insufficient support, while rules involving bin_4 are eliminated due to low lift.

Despite their rarity, some remaining rules show very high lift, suggesting that when these feature combinations occur, the probability of high article views is substantially higher than random expectation.

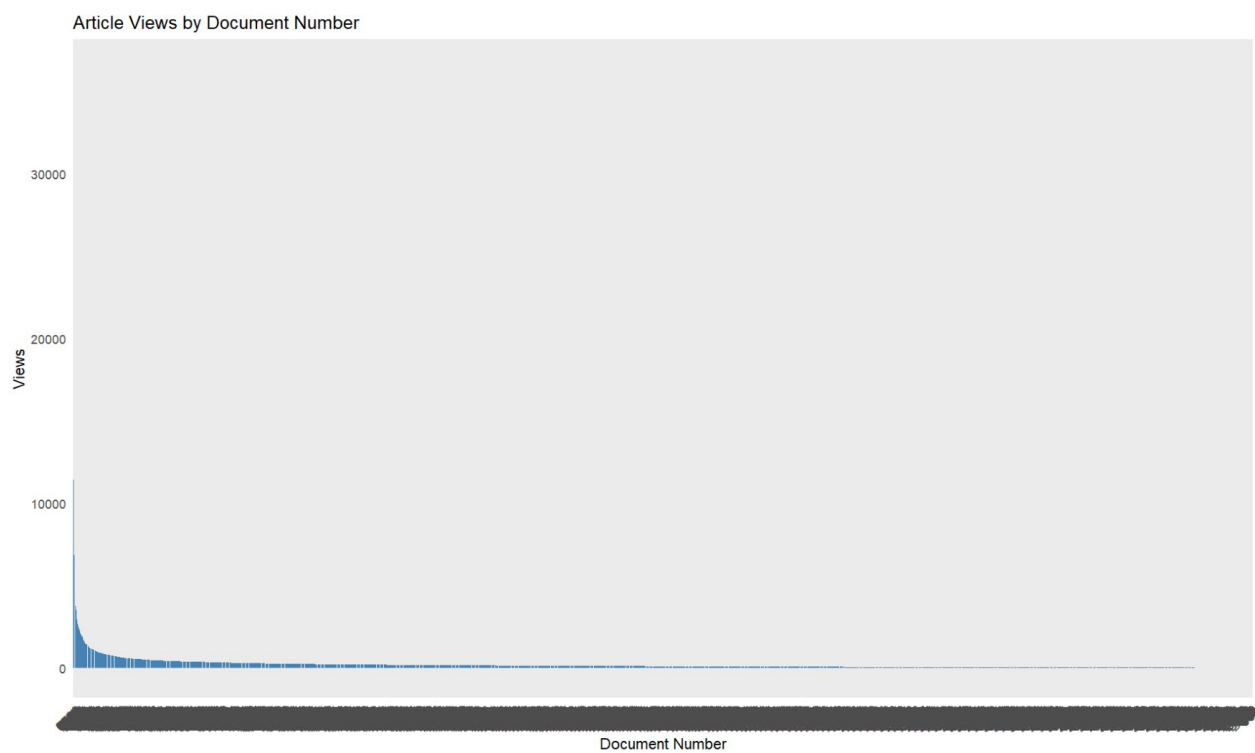


Figure 6.3.b Distribution of Views vs Papers

These findings indicate that predicting or improving article views based solely on pages, title, or abstract content is inherently difficult. Article views exhibit a pronounced Pareto effect: a small number of articles receive exceptionally high attention, while the majority receive relatively low attention.

7.Overall Summary

This study examines which article characteristics may influence article views. Under the constraint that publication year, authorship, and journal can't be modified, the analysis focuses on titles, abstracts, and article length.

1. A comprehensive preprocessing pipeline was implemented to standardize unstructured text, providing a reliable foundation for subsequent modeling.
2. LDA topic modeling was applied to titles and abstracts, converting text into quantifiable topic distributions. While dominant topics were generally weak, abstracts exhibited more discriminative topic structures than titles.
3. Bag of Words and TF-IDF analyses validated the LDA findings, confirming the prominence of terms such as *"method"* and *"model"* across the corpus.
4. K-means and hierarchical clustering were used to evaluate whether topic distributions could distinguish research directions. Although hierarchical clustering revealed clearer patterns, substantial overlap persisted, indicating that topic proportions alone are insufficient for stable classification or regression modeling.
5. Association analysis was then applied to identify co-occurring patterns of topic and page associated with high article views. While some rules exhibited high lift, their extremely low support and confidence highlight the sparsity of high-impact feature combinations.
6. Overall, article views follow a clear long-tail and Pareto distribution. Although certain title, abstract, or page characteristics may occasionally co-occur with high views, these features cannot be reliably used to predict or enhance article attention. This suggests that, in complex academic dissemination environments, article impact can't be determined solely by textual attributes.